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Paper Code: BBA(HM) 202 | Medical Record Science I | UPID: 2000082

Time: 3 Hours | Full Marks: 70 | Semester 2 — 2024–2025

Complete Answer Key — Exam Ready

GROUP-A | Very Short Answer Type | 1 × 10 = 10 Marks | Answer any TEN

QUESTION I

What is the full form of MR in healthcare?

ANSWER

MR = Medical Record. A Medical Record is a complete, written account of everything related to a patient's health — their past illnesses, current diagnosis, treatment given, and final outcome — all kept safely in the hospital for future reference.

QUESTION II

Name any two types of medical records.

ANSWER

Medical records are classified based on how and where the patient is receiving care in the hospital.

TWO TYPES

- **Inpatient Record —** For patients who are admitted and staying in the hospital (e.g., surgery case sheet, ward progress notes).
- **Outpatient Record —** For patients who visit the hospital for treatment but go back home the same day (e.g., OPD card, prescription).

QUESTION III

Name one method used for preserving medical records.

ANSWER

Microfilming is one of the most widely used methods for preserving medical records. In this process, paper documents are photographed at a very small (micro) size and stored on a film roll. This saves a huge amount of physical space and keeps records safe for over 100 years without fading or damage.

QUESTION IV

Mention one role of an MR technician.

ANSWER

One major role of an MR (Medical Record) Technician is **Medical Coding using the ICD system** (International Classification of Diseases). They read the doctor's diagnosis and convert it into a standard code number. These codes are used for hospital billing, insurance claims, and maintaining health statistics.

QUESTION V

What is meant by preservation of medical records?

ANSWER

Preservation of medical records means taking steps to protect patient documents from damage, decay, theft, or loss — so they stay readable and available for as long as required by law or hospital policy. This is done through methods like microfilming, digitization, fire-proof storage, and climate-controlled rooms.

QUESTION VI

Who is responsible for managing the Medical Records Department?

ANSWER

The **Medical Record Officer (MRO)** — also known as the **Health Information Manager (HIM)** — is the person in charge of the entire Medical Records Department. They manage all staff, ensure legal compliance, maintain systems, and oversee the proper storage and retrieval of all patient records.

QUESTION VII

Define medical audit.

ANSWER

A **Medical Audit** is a systematic review of patient care quality. It involves checking medical records against set standards to find out if the treatment provided was correct and complete — and then making improvements wherever there are gaps.

Simple way to remember: "Check → Compare → Improve."

QUESTION VIII

Name any one legal document maintained in medical records.

ANSWER

Informed Consent Form — This is a document signed by the patient (or their guardian) before any surgery or medical procedure. It shows that the patient was told about the risks involved and they agreed to go ahead. It is a very important legal protection for the hospital and the doctor.

QUESTION IX

What is one characteristic of a good medical record?

ANSWER

Accuracy is the most important characteristic of a good medical record. It means every detail written in the record — the patient's name, diagnosis, medicine dosage, treatment dates — must be correct and truthful. Even a small error can lead to wrong treatment or legal problems.

QUESTION X

Define retention of medical records.

ANSWER

Retention of medical records means keeping patient records stored safely for a fixed number of years as required by law or hospital policy — before they are destroyed or permanently archived.

Example: Adult records — 5 to 10 years | Records of minors — until they turn 18, plus extra years | MLC records — 10 or more years.

QUESTION XI

What is a medico-legal case?

ANSWER

A **Medico-Legal Case (MLC)** is a case where the patient's injury or illness is connected to the law — meaning the police or court is involved. In such cases, the hospital must inform the police and maintain all records very carefully as they can be used as evidence in court.

Examples: Road accidents, physical assault, poisoning, rape, burn cases, suspicious deaths.

QUESTION XII

Name one type of record involved in medico-legal cases.

ANSWER

Post-Mortem Report (Autopsy Report) — This report is written by a forensic doctor after examining a dead body. It clearly states the exact cause and manner of death. It is used as key evidence in criminal courts and during death investigations.

GROUP-B | Short Answer Type | 5 × 3 = 15 Marks | Answer any THREE

QUESTION 2

Describe the different types of medical records with examples. 5 Marks

INTRODUCTION

Medical records are official documents that store all information about a patient's health and treatment. Different situations in a hospital produce different types of records. Each type serves a specific purpose — clinical, legal, administrative, or diagnostic.

TYPES AT A GLANCE

- **Inpatient Record** — for admitted patients
- **Outpatient Record** — for OPD / day-visit patients
- **Emergency Record** — for casualty patients
- **Diagnostic Record** — lab and investigation reports
- **Medico-Legal Record** — for cases involving the law
- **Administrative Record** — birth, death, transfer documents

EXPLANATION OF EACH TYPE

1 **Inpatient Record**

Created when a patient is admitted to the hospital and stays in a ward. It contains admission notes, daily progress notes, doctor's orders, operation notes, and the discharge summary. *Example: Case sheet of a patient who had surgery.*

2 **Outpatient Record (OPD Record)**

Created for patients who visit the doctor for a check-up or treatment and leave on the same day without getting admitted. It includes the doctor's notes, prescriptions, and test reports. *Example: OPD card or folder.*

3 **Emergency / Casualty Record**

Maintained for patients who arrive at the hospital in an emergency — such as accident victims. It records the time of arrival, condition on arrival, immediate treatment given, and whether the case is medico-legal. *Example: Accident & Emergency register.*

4 **Diagnostic Record**

Contains results of all investigations done on the patient — blood tests, urine tests, X-rays, ECG, MRI, scan reports, and biopsy results. These help the doctor in making the correct diagnosis. *Example: Blood test report, Radiology report.*

5 **Medico-Legal Record**

Maintained for cases that have legal importance — such as assaults, rapes, poisoning, or suspicious deaths. These records are kept with extra security as they may be needed in court. *Example: MLC register, wound certificate, post-mortem report.*

6 **Administrative Record**

These are official documents maintained for government and institutional purposes. They include birth certificates, death certificates, transfer letters, and statistical reports. *Example: Death certificate issued by the hospital.*

CONCLUSION

Every type of medical record plays a unique role in patient care, hospital management, legal protection, and public health. Together, they form a complete picture of every patient's journey through the hospital.

QUESTION 3

Explain the importance of medical records in legal cases. **5 Marks**

INTRODUCTION

Medical records are not just health documents — they are powerful legal tools. In situations where a patient's injury or death involves the law, medical records become the most important piece of evidence. Courts, police, insurance companies, and lawyers all depend on these records to understand what happened and when.

KEY LEGAL ROLES OF MEDICAL RECORDS

- Evidence in court against or in favour of the doctor or hospital
- Official documentation in Medico-Legal Cases (MLC)
- Proof for insurance claim settlement
- Protection through consent forms
- Determination of cause of death
- Supporting patient's right to access their own records
- Establishing professional accountability of doctors

DETAILED EXPLANATION

1 Evidence in Court

If a patient or their family accuses a doctor of negligence or wrong treatment, the medical record is the strongest proof. A well-maintained record shows exactly what was done, when, and why — protecting the doctor from false accusations.

2 Medico-Legal Cases (MLC)

In cases of accidents, assaults, poisoning, or rape, the MLC record is the official document used by police and courts. It records the injuries, time of admission, and treatment given — all of which are critical in criminal investigations.

3 **Insurance Claim Settlement**

Insurance companies use medical records to verify whether the illness or injury actually happened, what treatment was given, and how long the patient was hospitalised — to process and settle the claim correctly.

4 **Consent Form as Legal Protection**

A signed consent form proves the patient agreed to the surgery or procedure after being fully informed of the risks. If the patient later complains, this form protects the hospital legally.

5 **Death Investigation**

In cases of suspicious or sudden death, the hospital records combined with the post-mortem report help the court determine whether the death was natural, accidental, or due to crime.

6 **Patient's Legal Right to Records**

By law, every patient has the right to get a certified copy of their own medical records. This is their legal right and the hospital is obligated to provide it upon request.

7 **Professional Accountability**

Records show who treated the patient, what decisions were made, and which medicines were given. This holds every doctor and nurse accountable for their clinical actions.

CONCLUSION

Medical records are a hospital's strongest legal shield. A well-written, accurate, and complete record protects doctors, helps patients get justice, supports insurance settlements, and ensures full legal accountability in the healthcare system.

QUESTION 4

Describe how MRD handles medico-legal documents. **5 Marks**

INTRODUCTION

Medico-Legal Cases (MLC) are among the most sensitive cases that arrive in a hospital — they involve the law, police, and courts. The Medical Records Department (MRD) handles these documents with extra care, following a strict step-by-step process to ensure accuracy, security, and legal compliance.

KEY STEPS IN HANDLING MLC DOCUMENTS

- Identification and tagging of MLC at entry
- Separate registration in the MLC Register
- Mandatory police intimation
- Secure and restricted storage

- Maintaining chain of custody for all document releases
- Long-term retention (10+ years)

FLOWCHART — MLC DOCUMENT HANDLING PROCESS

① Patient Identified as MLC	At Emergency / OPD entry, the doctor marks the case as MLC and attaches a special MLC tag to the file.
↓	
② MLC Registration	The case is entered in a dedicated MLC Register with a unique MLC number, date, time, and nature of injury.
↓	
③ Police Intimation	The police are informed immediately — by phone call and then in writing — as required by Indian law.
↓	
④ Complete Documentation	Every detail is recorded — injury description, time of arrival, treatment given, and name of treating doctor.
↓	
⑤ Secure Storage	MLC files are locked in restricted-access cabinets. Only authorised persons (MRO, treating doctor, police with written request) can access them.
↓	
⑥ Chain of Custody	Every time a document is given to police or court, it is recorded — who received it, date, time, purpose, and signature obtained.
↓	
⑦ Long-Term Retention	MLC records are kept for 10+ years because legal cases can continue for many years after the incident.

IMPORTANT POINTS EXPLAINED

1. **Why is police intimation mandatory?**

   Indian law requires hospitals to inform the police about any MLC case — even if the patient or family objects. This is a legal duty of the hospital, not a choice.

2. **Why is chain of custody important?**

   If MLC documents are mishandled or lost, it can damage the court case. The chain of custody ensures that every movement of the document is tracked so no one can claim the record was tampered with.

3. **Original records stay in MRD**

Even if police or court request the record, only a certified copy is given. The original never leaves the MRD.

CONCLUSION

The MRD plays a critical legal role in handling MLC documents. Proper identification, registration, secure storage, police intimation, and chain of custody ensure these records are legally sound and can be used reliably in court at any time.

QUESTION 5

Write a short note on assembling and deficiency checking of medical records. **5 Marks**

INTRODUCTION

After a patient is discharged from the hospital, their medical record is a collection of many loose papers from different departments — the ward, laboratory, radiology, pharmacy, and operation theatre. Two important MRD processes — **Assembling** and **Deficiency Checking** — bring all these papers together into one complete, correct, and usable medical record.

KEY POINTS

- Assembling: collecting and arranging all loose documents in standard order
- Deficiency Checking: checking the assembled record for missing or incomplete entries
- Deficiency slip sent to the doctor for completion
- Only complete records are permanently filed in MRD

FLOWCHART — ASSEMBLING & DEFICIENCY CHECKING PROCESS

Patient Discharged	All loose papers are collected from the ward, lab, radiology, OT, pharmacy, and nursing station.
↓	
Assembling	Papers are arranged in a fixed standard order: Face sheet → History & Examination → Doctor's Orders → Progress Notes → Lab Reports → Radiology → Consent Forms → Discharge Summary.
↓	
Deficiency Checking	Each document is checked. Common problems found: unsigned discharge summary, missing diagnosis code, incomplete history, missing consent form, illegible (unreadable) writing.
↓	

Deficiency Slip Issued

A deficiency slip is prepared and sent to the concerned doctor, nurse, or department to complete the missing part.

↓

Completion & Filing

Once all deficiencies are corrected, the complete record is sent for ICD coding and then permanently filed in MRD.

EXPLAINED IN DETAIL

1. **Why is assembling necessary?**
   Without assembling, papers from the ward are scattered and can be lost. Standard ordering makes it easy for any doctor or MRD staff to find any document instantly when needed later.

2. **Why is deficiency checking necessary?**
   An incomplete record is legally and medically dangerous. If a doctor's signature is missing on a discharge note, or a consent form is absent, the hospital has no protection if a legal problem arises later.

CONCLUSION

Assembling and deficiency checking together ensure that every patient record leaving the ward is complete, correctly ordered, and legally sound before it is permanently stored in the MRD. These are among the most important quality-control steps in medical records management.

QUESTION 6

Briefly describe the organizational structure of a Medical Records Department. 5 Marks

INTRODUCTION

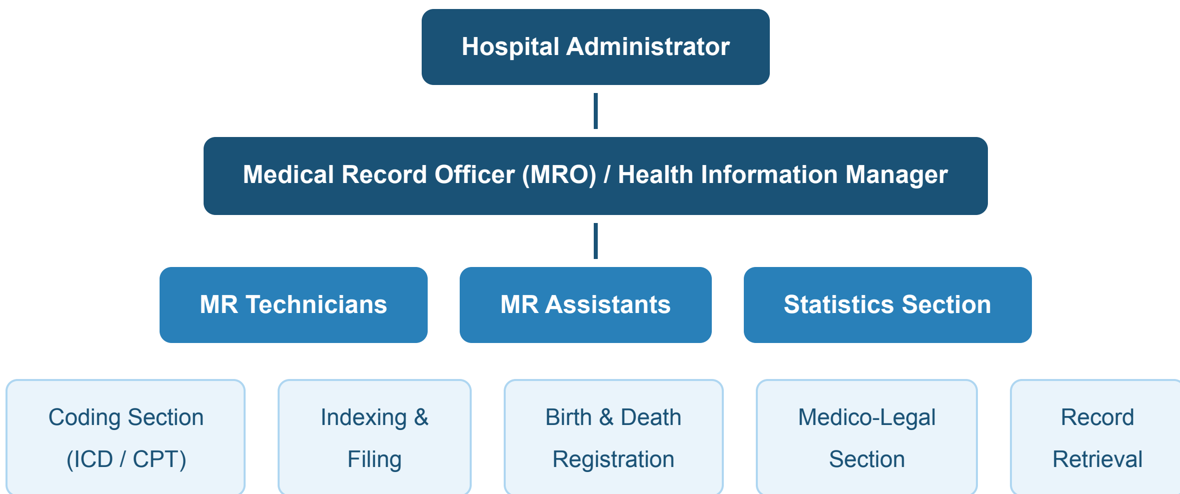
The Medical Records Department (MRD) is a well-organized unit in a hospital with different staff handling different responsibilities. It operates under the hospital administration and works alongside every clinical department. A clear hierarchy ensures that all functions — from coding to MLC to statistics — are handled by the right person.

KEY SECTIONS IN AN MRD

- Medical Record Officer (MRO) — overall head
- Medical Record Technicians — coding, indexing, retrieval
- Medical Record Assistants — filing, assembling, deficiency checking
- Coding Section — ICD/CPT coding for billing
- Statistics Section — hospital data and reports

- Birth & Death Registration — official government documents
- Medico-Legal Section — MLC handling

ORGANIZATIONAL CHART — MRD



ROLE OF EACH SECTION

1. **Medical Record Officer (MRO)**
   Head of the entire department. Responsible for managing staff, ensuring legal compliance, maintaining systems, and liaising with hospital administration and clinical departments.

2. **Medical Record Technicians**
   Handle ICD coding of diagnoses and procedures, indexing of records, and retrieval of patient files when requested by doctors or the court.

3. **Medical Record Assistants**
   Handle the day-to-day physical work — collecting records from wards after discharge, assembling, deficiency checking, and filing in the storage room.

4. **Statistics Section**
   Compiles important hospital data — bed occupancy rate, average length of stay, death rates, OPD attendance — and submits reports to management and government authorities.

5. **Medico-Legal Section**
   Specially handles all MLC files — maintains the MLC register, communicates with police, and ensures strict confidentiality and long-term safe storage of these sensitive documents.

CONCLUSION

A well-structured MRD with clearly defined roles ensures that every record is correctly created, stored, coded, and retrieved. It is the information hub of the hospital — connecting all departments and supporting clinical care, legal compliance, research, and administration.

GROUP-C | Long Answer Type | 15 × 3 = 45 Marks | Answer any THREE

QUESTION 7

Explain various methods used in the preservation of medical records, both manual and electronic.

15 Marks

INTRODUCTION

Medical records contain vital information about a patient's health — and they must be kept safe, readable, and available for years, sometimes even decades. **Preservation** refers to all the steps taken to protect medical records from physical damage (fire, water, insects, decay) and from loss or unauthorized access. Methods can be broadly divided into **manual (physical)** and **electronic (digital)** methods.

METHODS AT A GLANCE

- **Manual:** Microfilming, Microfiching, Lamination, Acid-free folders, Fire-proof storage, Climate control, Pest control
- **Electronic:** EHR, Scanning & Digitization, Cloud Storage, RAID, Optical Discs, Encrypted Backup, Disaster Recovery

FLOWCHART — CHOOSING A PRESERVATION METHOD

Medical Record Created — Patient discharged; all documents assembled and checked for deficiencies.

↓

Is the hospital fully digital? — If YES → Use EHR / Cloud storage. If NO → Use Manual + Hybrid methods.

↓

Manual Method Selected — Microfilming, Acid-free folders, Fire-proof cabinets, Climate-controlled room.

↓

Electronic Method Selected — Scanning, EHR entry, Cloud upload, Encrypted backup, RAID storage.

↓

Preserved & Indexed	Record stored with index number for fast retrieval when needed in future.

MANUAL METHODS — DETAILED EXPLANATION

1. **Microfilming**

   Paper records are photographed at a tiny (micro) scale and stored on a film roll. A single roll can store thousands of documents. Films can last 100+ years and are ideal for hospitals with huge volumes of old records.

2. **Microfiching**

   Similar to microfilming, but images are stored on flat plastic sheets called microfiche cards. These are easy to view on a microfiche reader machine and are good for storing index cards and summary documents.

3. **Lamination**

   Individual important documents (like consent forms or certificates) are sealed in a thin plastic cover to protect them from moisture, tears, and frequent handling. Used for documents that are often referred to.

4. **Acid-free Folders**

   Normal paper and folders contain acid that slowly destroys documents over years. Acid-free, archival-quality folders do not release any harmful chemicals, making them ideal for long-term storage of original records.

5. **Fire-proof & Climate-controlled Storage**

   Records are stored in steel fire-proof cabinets in rooms where temperature and humidity are controlled. This prevents damage from fire, flood, excessive heat, or dampness. Pest control is also done regularly to prevent insects from eating paper records.

ELECTRONIC METHODS — DETAILED EXPLANATION

1. **Electronic Health Records (EHR)**

   All patient information is entered directly into a computer system — no paper needed. It is the most modern and efficient method. Records can be retrieved instantly by any authorised person from any location.

2. **Scanning & Digitization**

   Existing paper records are scanned using a high-resolution scanner and saved as PDF or TIFF image files on a computer server. This creates a digital backup of physical records.

3. **Cloud Storage**

   Digital records are uploaded to secure internet-based servers. They can be accessed from anywhere with proper login credentials. Cloud systems automatically back up data so that nothing is lost even if the hospital's local server crashes.

4. **RAID Systems**

   RAID stands for Redundant Array of Independent Disks. Multiple hard drives store the same data simultaneously. If one drive fails, the others keep the data safe. This is used in hospitals that manage large volumes of digital records.

5 **Encrypted Backup & Disaster Recovery**

All digital data is encrypted (locked with a code) to prevent hacking. Automated backup copies are created regularly and stored at a different physical location. In case of fire, flood, or cyberattack, the disaster recovery system restores everything quickly.

DIFFERENCE — MANUAL VS ELECTRONIC METHODS

Basis of Difference	Manual Methods	Electronic Methods
Storage Space	Requires large physical space (cabinets, rooms)	Very compact — thousands of records fit on one server
Retrieval Speed	Slow — staff must manually search through files	Instant — searchable by patient name, ID, or date
Security	Lock & key, fire-proof cabinets	Password, encryption, firewall, audit trails
Risk of Damage	Fire, flood, pests, ageing, decay	Hacking, power failure (controlled by backup systems)
Cost	Low setup cost, but high space and maintenance cost	Higher setup cost, but very low running cost over time
Access	Only one person can access at a time (original file)	Multiple authorised users can access simultaneously
Best Suited For	Small or rural hospitals with limited technology	Large, modern hospitals and multi-branch healthcare systems

CONCLUSION

No single method is perfect on its own. The best approach for any hospital is a **combination** of both — keeping physical records in proper conditions while also maintaining digital backups. This ensures records are always available, safe from every type of risk, and accessible when needed for patient care, legal purposes, or research.

QUESTION 8(A)

Elaborate on the types, process, and outcomes of medical audits. Why are they essential in hospitals?

10 Marks

INTRODUCTION

A **Medical Audit** is a formal process of reviewing and evaluating the quality of patient care in a hospital. It involves examining medical records and comparing the care provided against established medical standards. The goal is simple — to find out *what went wrong, what went right, and how things can be improved* to make the hospital safer and better for every patient.

KEY POINTS

- Types: Prospective, Concurrent, Retrospective, Internal, External
- Process: 7-step audit cycle — choose topic → set standards → collect data → compare → find gaps → make changes → re-audit
- Outcomes: Improved care, fewer errors, better documentation, accountability, accreditation
- Essential because: patient safety, legal compliance, quality certification (NABH/JCI)

THE AUDIT CYCLE — STEP BY STEP PROCESS

Step 1 — Choose a Topic

Select a specific aspect of care to review. Example: "Are all patients with fever getting a blood test within 2 hours?"

↓

Step 2 — Set Standards

Define what "good care" looks like. Example: "90% of fever patients should get blood test within 2 hours."

↓

Step 3 — Collect Data

MRD staff review medical records of relevant patients and record the data systematically.

↓

Step 4 — Compare Results

Compare what was actually done against the standard set. Example: "Only 60% got the test in time — there is a gap."

↓

Step 5 — Identify Gaps

Find out why the gap exists — was it a staffing problem, a communication failure, or a lack of equipment?

↓

Step 6 — Implement Changes

Make the necessary improvements — new training, new protocols, or more staff at peak hours.

↓

Step 7 — Re-Audit

After changes are made, repeat the audit to check if the improvement actually happened. Then the cycle begins again.

TYPES OF MEDICAL AUDIT — EXPLAINED

1. **Prospective Audit**

   Done *before* treatment begins. The patient's condition is reviewed to plan the best possible care in advance. Useful in planned surgeries or for high-risk patients.

2. **Concurrent Audit**

   Done *while* the patient is still in the hospital receiving care. If a problem is found, it can be corrected immediately — before any harm is done. Very effective for ICU or long-stay patients.

3. **Retrospective Audit**

   Done *after* the patient is discharged. The most common type. It identifies patterns over many cases and helps the hospital improve its overall standard of care.

4. **Internal Audit**

   Conducted by the hospital's own doctors, nurses, or MRD team. It is done regularly as an in-house quality check without involving outside agencies.

5. **External Audit**

   Conducted by outside experts or accreditation bodies like NABH (National Accreditation Board for Hospitals) or JCI (Joint Commission International). This is done as part of the hospital's quality certification process.

OUTCOMES & WHY AUDITS ARE ESSENTIAL

- ✓ **Improved Quality of Patient Care**

  When doctors and nurses know their work will be reviewed, they follow correct procedures more carefully — leading to better outcomes for patients.

- ✓ **Reduction in Medical Errors**

  Audits identify recurring mistakes — wrong medicines, missed diagnoses, delayed treatments — and fix the root cause so they don't happen again.

- ✓ **Better Record Keeping**

  When auditors check records, doctors realize the importance of complete and accurate documentation — improving the overall quality of medical records in the hospital.

- ✓ **Hospital Accreditation (NABH / JCI)**

  Regular medical audits are a mandatory requirement for hospital accreditation. A hospital with strong audit systems gets certified as a quality healthcare provider — attracting more patients and better insurance empanelment.

CONCLUSION

Medical audits are not about finding fault — they are about finding opportunities to improve. A hospital that audits itself regularly becomes safer, more efficient, and more trustworthy. The audit cycle ensures that improvement is not a one-time event, but a continuous process that never stops.

QUESTION 8(B)

What are the guidelines for determining staffing levels and requirements in the MRD?

5 Marks

INTRODUCTION

The MRD cannot function well without the right number of trained staff. Too few staff means delays, errors, and poor record management. Too many staff is a waste of resources. Hospital administrators use specific guidelines and factors to calculate exactly how many MRD staff are needed.

KEY FACTORS FOR STAFFING

- Number of hospital beds
- Daily patient load (OPD + IPD)
- Type of hospital (teaching / specialty / general)
- Complexity of services offered
- Required turnaround time for records
- Degree of computerization in MRD

EACH FACTOR EXPLAINED

1 **Number of Hospital Beds**

More beds means more admitted patients, more records, more coding, more filing. The general standard is **1 MR Technician per 50 to 100 beds**.

2 **Daily Patient Load (OPD + IPD)**

A hospital seeing 1,000 OPD patients daily needs more staff than one seeing 200. High patient volume means more records are created, assembled, coded, and retrieved every day.

3 **Type of Hospital**

A **teaching hospital** needs more MRD staff because records are used not just for patient care, but also for student training, research, and publishing case studies. A **specialty hospital** needs expert coders for complex procedures.

4 **Turnaround Time Required**

If hospital policy says all records must be coded and filed within 24 hours of discharge, more staff will be needed to meet this deadline compared to a hospital with a 72-hour policy.

5 Degree of Computerization

In a fully digital hospital with EHR, many manual tasks are automated — so fewer staff are needed for physical filing and retrieval. But more skilled IT-trained staff are needed for data entry and system management.

CONCLUSION

Staffing in MRD is not a one-size-fits-all formula. It must be calculated based on the hospital's size, workload, type, and technology level. The right staffing ensures records are managed efficiently, legally, and without errors.

QUESTION 9

How do MRD personnel ensure legal compliance and protect patient rights in the handling of records?

15 Marks

INTRODUCTION

The MRD holds the most sensitive information that exists in a hospital — every patient's personal health data. This comes with two major responsibilities. First, the MRD must follow all **legal rules** about how records are stored, released, and destroyed. Second, it must actively **protect the rights of every patient** — their right to privacy, their right to access their records, and their right to have accurate information maintained about them. Both responsibilities go hand in hand.

KEY POINTS

- **Legal Compliance:** Retention policy, authorised access only, MLC protocol, accurate ICD coding, proper consent filing, birth/death registration
- **Patient Rights:** Right to confidentiality, right to access own records, right to correction, data security, audit trails, staff training

FLOWCHART — LEGAL PROCESS FOR RELEASING A MEDICAL RECORD

Request Received	Request comes from: patient, treating doctor, police (for MLC), court, or insurance company.
↓	
Verify Identity & Authority	Check: Is this person authorised to receive this record? (ID proof, court order, or valid consent from patient required.)
↓	

MRO Approval	The Medical Record Officer reviews and formally approves the release of the record.
↓	
Certified Copy Prepared	A certified photocopy is made. The ORIGINAL record always stays in the MRD — it never leaves.
↓	
Entry in Release Register	All details recorded: who received it, date, time, purpose, and their signature.

LEGAL COMPLIANCE — EXPLAINED

1. **Retention Policy**

   Records are kept for the legally required period before disposal — Adult records: 5 to 10 years | Records of minors: until they reach 18 years + additional years | MLC records: 10+ years. Destroying a record too early is a legal offence.

2. **Authorised Access Only**

   Records can only be given to the treating doctor, the patient themselves, a court of law with written order, or an insurance company with valid patient consent. Random visitors, family members (without consent), or press are never given access.

3. **MLC Protocol Compliance**

   For all medico-legal cases, strict protocols are followed — police intimation within the correct time frame, separate MLC register, restricted access, and detailed documentation of all information related to the case.

4. **Accurate ICD Coding**

   Incorrect codes can lead to wrong billing, insurance fraud, or legal disputes. MRD ensures every diagnosis and procedure is correctly coded using ICD-10/11 as per official guidelines.

PATIENT RIGHTS — EXPLAINED

1. **Right to Confidentiality**

   A patient's health information is private. MRD staff are ethically and legally bound to never share patient details with any unauthorised person — not even the patient's employer or relatives without the patient's consent.

2. **Right to Access Own Records**

   Every patient has the legal right to receive a certified copy of their own medical record at any time upon request. The hospital must provide this within a reasonable time — usually within 48 to 72 hours.

3. **Right to Correction of Errors**

   If a patient finds an error in their medical record — a wrong date, incorrect diagnosis, or wrong medication — they can formally request a correction. The MRD investigates and corrects it with proper documentation.

4. **Data Security for Digital Records**

Electronic records are protected by passwords, role-based access (each staff member can only see what their role allows), end-to-end encryption, and firewall systems to prevent hacking or data leaks.

5 **Audit Trails**

Every time a digital record is opened, edited, or downloaded, the system automatically records who did it and when. This makes it impossible to access records secretly without being caught.

DIFFERENCE — LEGAL COMPLIANCE VS PATIENT RIGHTS PROTECTION

Basis	Legal Compliance	Patient Rights Protection
Focus	Follows rules set by the government and law	Protects the individual patient's personal interests
Who benefits	Hospital, government, court system	The patient directly
Key actions	Retention, MLC protocol, authorized access, coding	Confidentiality, right to access, data security, correction
Consequence of failure	Legal action against the hospital (fines, license issues)	Breach of patient trust, ethical violations, patient harm

CONCLUSION

Legal compliance and patient rights protection are two sides of the same coin in MRD. Following the law protects the hospital, while protecting patient rights builds trust and ensures ethical healthcare. Together, they ensure that every patient record is handled with the highest level of responsibility, security, and respect.

QUESTION 10

How can the Medical Records Department ensure the accuracy, confidentiality, and timely availability of patient records while adapting to digital transformation and increasing patient volumes?

15 Marks

INTRODUCTION

Modern hospitals are growing rapidly — more patients, more records, and more digital systems. The MRD must rise to this challenge without ever compromising on three core pillars: **Accuracy** (records must be correct), **Confidentiality** (records must be private), and **Timely Availability** (records must be accessible when needed). As hospitals go digital, new tools and strategies make all three pillars stronger — but only if managed properly.

KEY POINTS

- **Accuracy:** EHR validation, deficiency checking, ICD coding standards, staff training, periodic audits
- **Confidentiality:** Role-based access, encryption, multi-factor login, cybersecurity audits, strict policies
- **Timely Availability:** Cloud access, barcode/RFID tagging, 24/7 digital systems, disaster recovery
- **Managing Volume:** Scalable cloud infrastructure, automation, workflow management, more trained staff

FLOWCHART — DIGITAL MRD IN ACTION

Patient Admitted	Patient registered in EHR system — no paper needed. Unique digital ID assigned.
↓	
All Clinical Data Entered Digitally	Doctors, nurses, lab, radiology all enter data into the same EHR — real time, accurate, and accessible to all authorised staff.
↓	
Validation & Accuracy Checks	EHR system alerts if data is missing, incorrectly formatted, or contradicts earlier entries — preventing errors automatically.
↓	
Discharge & Coding	At discharge, MR Technician assigns ICD-10/11 codes. System checks coding accuracy before finalising the record.
↓	
Encrypted Storage & Cloud Backup	Completed record is stored with encryption in the hospital server and automatically backed up to a secure cloud location.
↓	
Access on Demand — Securely	Authorised doctor, patient, or court can access the record instantly — with login, role-based access, and a full audit trail of who viewed it.

THREE PILLARS — DETAILED EXPLANATION

1 Ensuring Accuracy

EHR systems with built-in validation checks automatically flag missing fields or wrong formats. Regular deficiency checking ensures no discharge record is incomplete. ICD-10/11 coding standards are followed strictly. Periodic medical audits catch patterns of errors and correct them. All staff are trained regularly on documentation standards.

2 Ensuring Confidentiality

Role-based access control ensures each staff member can only view the records relevant to their job — a receptionist cannot see clinical notes. All data is encrypted end-to-end. Multi-factor login (password + OTP) prevents unauthorised access. Regular cybersecurity audits detect and fix vulnerabilities before they are exploited.

3 Ensuring Timely Availability

Cloud-based systems make records available from any department, or even from another hospital branch, instantly. Barcode and RFID tagging on physical files allow quick location and retrieval. Emergency teams get 24/7 digital access to critical patient history. Disaster recovery systems ensure the hospital can access all records even after a server failure or natural disaster.

4 Managing Increasing Patient Volumes

Scalable cloud infrastructure automatically increases storage capacity as patient numbers grow — no physical expansion needed. Automation of routine tasks like appointment records, discharge summaries, and report generation reduces the manual workload on staff. Workflow management software distributes tasks evenly among MRD staff so no one is overloaded.

DIFFERENCE — TRADITIONAL MRD VS DIGITAL MRD

Basis	Traditional / Paper MRD	Digital / EHR-based MRD
Record Creation	Written manually by doctors and nurses	Entered directly into computer / tablet in real time
Storage	Physical files in large storage rooms	Servers and cloud — no physical space needed
Retrieval	Manual search — can take hours	Instant — search by name, ID, or date
Accuracy	Prone to human errors, illegible writing	Validation checks prevent most errors automatically
Confidentiality	Lock and key, limited to physical location	Encryption, passwords, audit trails, firewalls
Handling Volume	More patients = more physical staff needed	Scalable — system handles more records automatically
Disaster Risk	Fire or flood can destroy everything	Cloud backup keeps data safe even in disasters

CONCLUSION

Digital transformation is not optional anymore for modern MRDs — it is a necessity. By adopting EHR systems, cloud storage, encryption, and smart workflow tools, the MRD can simultaneously be more accurate, more private, and faster — even as hospitals grow bigger and busier. The goal is a paperless, people-friendly, and legally sound record management system.

QUESTION 11

Discuss the role of medical records in healthcare delivery. Include a flow chart of their function.

15 Marks

INTRODUCTION

Medical records are the foundation of healthcare delivery. Every single decision made in a hospital — from the medicine a doctor prescribes to the bill a patient pays — is based on what is written in the medical record. Without accurate and complete medical records, safe and effective healthcare is simply not possible. A medical record follows the patient from the moment they arrive at the hospital to long after they leave — connecting every doctor, nurse, lab, and department involved in their care.

KEY ROLES OF MEDICAL RECORDS IN HEALTHCARE

- Supports clinical decision-making and diagnosis
- Ensures continuity of care across multiple doctors and visits
- Acts as a communication tool between all healthcare teams
- Provides legal protection for patients and hospitals
- Supports medical research and education
- Enables accurate billing and insurance processing
- Forms the basis of medical audits and quality assurance
- Helps in public health monitoring and disease surveillance

FLOWCHART — COMPLETE JOURNEY OF A MEDICAL RECORD

🏥 **Patient Arrives at Hospital**	Patient visits OPD, Emergency, or comes for admission. Journey of the record begins here.

↓

📝 **Registration**	New patient → Unique ID assigned → OPD card or IP record created → Basic details entered.

↓

🩺 **Clinical Consultation**	Doctor records: Chief complaint, medical history, physical examination findings, provisional diagnosis, and treatment plan.

↓

🔬 **Investigations**	Lab tests, X-rays, ECG, MRI, scan reports are done and results added to the patient's record.
↓	
💊 **Treatment / Surgery**	Daily progress notes, doctor's orders, drug charts, nursing notes, and operation notes are recorded throughout the stay.
↓	
🏠 **Discharge**	Discharge summary prepared — final diagnosis, treatment given, condition at discharge, follow-up instructions, and medicines prescribed.
↓	
📁 **MRD — Assembling**	All loose documents collected from ward and arranged in standard order in MRD.
↓	
🔍 **Deficiency Checking**	Missing signatures, reports, or entries identified → Deficiency slip sent to doctor for completion.
↓	
🏷 **ICD Coding**	MR Technician assigns standard ICD-10/11 codes to the diagnosis and all procedures performed.
↓	
🗄 **Indexing & Filing**	Record added to disease index, physician index, patient index → Permanently filed using terminal digit or serial filing system.
↓	
🔄 **Retrieval & Multiple Uses**	Record retrieved as needed for: follow-up care | legal purpose | insurance | research | medical audit | public health statistics.

EACH ROLE EXPLAINED IN DETAIL

1 Clinical Care and Diagnosis

A doctor uses the patient's past records to know what illnesses they have had, what medicines they are allergic to, and what treatments have been tried before. This prevents dangerous mistakes like prescribing a drug the patient is allergic to — and ensures the right treatment is given from day one.

2 Continuity of Care

When a patient visits multiple doctors — a general physician, a specialist, a surgeon — each doctor can refer to the same record and give consistent care. Without this, every doctor would start from scratch, wasting time and risking contradictory treatments.

3 Communication Between Teams

Doctors, nurses, lab technicians, pharmacists, and physiotherapists all refer to the same medical record. It is the single shared source of truth that keeps everyone on the same page and prevents miscommunication.

4 **Legal Protection**

Records are evidence in court for medico-legal cases, medical negligence claims, and insurance disputes. A well-maintained record protects both the patient and the hospital.

5 **Research and Medical Education**

Researchers use anonymised medical records to study disease patterns, treatment success rates, and drug reactions. Medical students and interns learn from real case records — making records an essential part of training the next generation of doctors.

6 **Billing, Insurance, and Administration**

The ICD codes assigned from the medical record are used to generate accurate hospital bills and process insurance claims. Without correct records, billing errors and insurance rejections are inevitable.

7 **Quality Assurance and Medical Audits**

Medical audits use records to check if the right care was given. Records reveal patterns — if one ward has more post-surgery infections than others, the audit helps identify why and fix it.

8 **Public Health and Disease Surveillance**

Government health departments use hospital records to track outbreaks of diseases (like dengue or COVID-19), monitor birth and death rates, plan vaccination drives, and allocate healthcare resources to the right areas.

CONCLUSION

Medical records are far more than just paperwork — they are the living memory of every patient's healthcare journey. They connect every department, every doctor, and every decision in a hospital into one coherent, accountable, and safe system. A hospital that maintains excellent medical records delivers excellent healthcare. The medical record is truly the backbone of every healthcare system in the world.